Lucas Barnes

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Education

Colorado State University — Fort Collins, CO
B.S. Mechanical Engineering · GPA: 3.67 — *Aug 2024 – May 2027 (Expected)*

Cabrillo College — Santa Cruz, CA
A.S. Mechanical Engineering · GPA: 3.3 — *Aug 2022 – May 2024*

Technical Skills

Design & Analysis: SolidWorks · GD&T · MATLAB · Design for Manufacturing & Assembly (DFMA) · FEA · Fatigue & Structural Analysis · Thermal Modeling & Testing · Tolerance Analysis & Fits

Mechatronics & Test: LabVIEW · Arduino · Sensor Integration · Closed-Loop Control · C/C++ · oscilloscopes · function generators · Data Analysis

Manufacturing: CNC Machining (HAAS Mill & Lathe) · Fiber-Optic Splicing · Manual Mill & Lathe · Injection Molding · 3D Printing (Rapid Prototyping) · Laser 3D Scanning · Vat Photopolymerization (VPP) · Laser Cutting · Forging · Mechanical Assembly · Soldering

Documentation: BOM & Cost Estimation · SOPs · Work Instructions · Acceptance Test Procedures (ATP) · Operation and Maintenance Manuals · Engineering Drawings

Projects

Sierra Space Capstone – Solar Array Thermal Test Rack
Lead Design Engineer — CAD, Modular Architecture, Chamber Fit, Documentation

- Leading the design of a modular test rack for thermal cycling between ($-150\,^{\circ}$C to $+150\,^{\circ}$C) to fit nine different current solar array programs as well as future array designs and projecting 4–6x more solar array wings tested per chamber run
- Writing and tracking requirements, running trade studies for concept generation, and documenting design decisions
- Producing 3D models, drawings, assembly documentation, operation and maintenance manual (O&M), and acceptance test procedure (ATP)

Manufacturing & Test Engineer — DFMA, Procurement, Build

- Will lead the design for manufacturing & assembly (DFMA) reviews and manufacturability assessments, develop bill of materials (BOM), procurement strategy, vendor quote packages, and document places for improvement

Gear Shaft Design Project

- Designed a power-transmission gear shaft for a consumer-vehicle gearbox (80 HP at 600 rpm; 8,400 lbf-in torque, 4,900 lbf-in max bending moment) in 1020 HR steel, meeting the required $\geq$2.5 factor of safety at every critical section under combined cyclic loading
- Performed fatigue and yield stress analysis, by hand calculation, MATLAB, and FEA at six stress concentrations
- Selected a bearing and specified keyways, retaining rings, a locational-interference fit, and dimensioned them along with the shaft in SolidWorks

Freezer Beverage Cooling Model

- Built and solved a transient thermal model of a can in a $-18\,^{\circ}$C freezer, combining forced and natural convection, radiation and shelf conduction in MATLAB
- Validated against a physical test ($4.0\,^{\circ}$C predicted, $9.9\,^{\circ}$C measured at 40.5 min) and traced the error to estimated airflow and fan duty cycle

Manufacturing Projects

- Authored SOPs and work instructions for mill, lathe, and CNC operations, forming the documentation basis for build, validation, and acceptance testing
- Manufactured functional parts to drawing specifications across manual mill/lathe, CNC mill/lathe, injection molding, and forging processes

Timed Pill Dispenser

- Built a water-bottle-mounted dispenser for 4 medications at 16 set times integrating IR sensors, servos, an OLED screen and a RTC

Experience

Colorado State University, Dept. of Mechanical Engineering — Fort Collins, CO
Undergraduate Teaching Lab Assistant (TA), MECH 207 & MECH 231 — *Aug 2026 – Present*

- Support the MECH 207 (mechatronics) and MECH 231 (engineering experimentation) teaching labs: preparing labs and troubleshooting one-on-one with students
- Coach students through Arduino-based electro-mechanical builds, lab instrumentation (oscilloscopes, function generators), and experimental data analysis and measurement uncertainty with LabVIEW

Colorado State University Parking & Transportation — Fort Collins, CO
Student Service Advisor — *Aug 2024 – Present*

- Facilitate university-wide parking and transportation services including vehicle reservations, cleaning, and billing

Bicycle Trip — Santa Cruz, CA
Bike Assembler — *Jul 2020 – Jan 2023*

- Assembled and tuned new mountain and road bikes to manufacturer specifications, and repaired customer bikes (brake bleeds, bottom-bracket replacements, and drivetrain tune-ups) to safe riding condition

Leadership & Certifications

- University Technology Fee Advisory Board (UTFAB), Voting Member at Large: help allocate a $2.5M annual tech fee across campus proposals
- HAAS CNC Vertical Mill Certification: set up and operate a HAAS CNC vertical mill (December 2024)
- CSU Baja SAE Design Team, Member: configured the competition vehicle's full SolidWorks assembly
- Dean's List: Fall 2025